Laplace Transform Theory

1 Introduction

Let

$$\Psi_b(t) = \Psi_{b\,0} \left\{ \begin{array}{ll} 0 & t < 0 \\ 1 & t > 0 \end{array} \right. . \tag{1}$$

It follows that

$$\bar{\Psi}_b(g) = \Psi_{b\,0} \int_0^\infty \mathrm{e}^{-g\,t}\,dt = \frac{\Psi_{b\,0}}{g}. \tag{2}$$

Now, the Laplace transformed reconnected magnetic flux is

$$\bar{\Psi}_0(g) = \frac{E_{sb}\,F_s(\hat{g})\,\bar{\Psi}_b(g)}{S^{1/3}\,\hat{\Delta}_s(\hat{g}) - E_{ss}} = E_{sb}\,\Psi_{b\,0}\,\frac{1}{g}\,\frac{F_s(\hat{g})}{S^{1/3}\,\hat{\Delta}_s(\hat{g}) - E_{ss}}, \tag{3}$$

where $\hat{g} = S^{\,1/3}\,g\,\tau_H$. The reconnected flux is

$$\Psi_0(t) = \frac{1}{2\pi\,\mathrm{i}} \int_c \bar{\Psi}_0(g)\,\mathrm{e}^{\,g\,t}\,dg, \tag{4}$$

where c is the Bromwich contour. Thus, we obtain

$$\Psi_0(\hat{t}) = \frac{1}{2\pi\,\mathrm{i}} \int_c \bar{\Psi}_0(\hat{g})\,\mathrm{e}^{\,\hat{g}\,\hat{t}}\,d\hat{g}, \tag{5}$$

where $\hat{t} = t/(S^{\,1/3}\,\tau_H)$ and

$$\bar{\Psi}_0(\hat{g}) = E_{sb}\,\Psi_{b\,0}\,\frac{1}{\hat{g}}\,\frac{F_s(\hat{g})}{S^{1/3}\,\hat{\Delta}_s(\hat{g}) - E_{ss}}. \tag{6}$$

2 Non-Constant-ψ Regimes

We shall assume that $|S^{1/3}\,\hat{\Delta}_s| \gg |E_{ss}|$ in the non-constant-ψ regimes, so

$$\bar{\Psi}_0(\hat{g}) \simeq \frac{E_{sb}\,\Psi_{b\,0}}{S^{1/3}}\,\frac{1}{\hat{g}}\,\frac{F_s(\hat{g})}{\hat{\Delta}_s(\hat{g})}. \tag{7}$$

2.1 Inertial Regime

In the inertial regime,

$$\hat{\Delta}_s(\hat{g}) = -\frac{\pi}{(\hat{g}+\mathrm{i}\,Q_E)^{1/2}\,[\hat{g}+\mathrm{i}\,(Q_E+Q_i)]^{1/2}}, \tag{8}$$

$$F_s(\hat{g}) = -\frac{2}{[\hat{g}+\mathrm{i}\,(Q_E+Q_e)]\,(\hat{g}+\mathrm{i}\,Q_E)\,[\hat{g}+\mathrm{i}\,(Q_E+Q_i)]}, \tag{9}$$

$$\frac{F_s(\hat{g})}{\hat{\Delta}_s(\hat{g})} = \frac{2}{\pi}\,\frac{1}{[\hat{g}+\mathrm{i}\,(Q_E+Q_e)]\,(\hat{g}+\mathrm{i}\,Q_E)^{1/2}\,[\hat{g}+\mathrm{i}\,(Q_E+Q_i)]^{1/2}} \tag{10}$$

Thus,

$$\begin{aligned}\bar{\Psi}_0(\hat{g}) &= \frac{\Psi_\infty\,\epsilon}{\mathrm{i}\,(Q_E+Q_e)}\left[\frac{1}{\hat{g}} - \frac{1}{\hat{g}+\mathrm{i}\,(Q_E+Q_e)}\right]\\ &\quad\times\frac{1}{(\hat{g}+\mathrm{i}\,Q_E)^{1/2}\,[\hat{g}+\mathrm{i}\,(Q_E+Q_i)]^{1/2}},\end{aligned} \tag{11}$$

where

$$\epsilon = \frac{2}{\pi}\,\frac{(-E_{ss})}{S^{1/3}}, \tag{12}$$

$$\Psi_\infty = \frac{E_{sb}\,\Psi_{b\,0}}{(-E_{ss})}. \tag{13}$$

Under normal circumstances $0 < \epsilon \ll 1$. Moreover, Ψ_∞ is the final amount of driven reconnection in the absence of suppression.

Now, if

$$\bar{\Psi}_0(\hat{g}) = \frac{1}{\hat{g}+a}\,\bar{H}(\hat{g}) \tag{14}$$

then (ERD:[1] 4.1.20)

$$\Psi_0(\hat{t}) = \int_0^{\hat{t}} \mathrm{e}^{\,a\,(\hat{t}'-\hat{t})}\,H(\hat{t}')\,d\hat{t}'. \tag{15}$$

[1]Erdélyi, et al. *Tables of Integral Transforms*, vol. I.

In this case,

$$\bar{H}(\hat{g}) = \frac{1}{(\hat{g} + \mathrm{i}\,Q_E)^{1/2}\,[\hat{g} + \mathrm{i}\,(Q_E + Q_i)]^{1/2}} \tag{16}$$

$$= \frac{1}{(\hat{g}^{\,2} + \alpha\,\hat{g} + \beta)^{1/2}}, \tag{17}$$

where

$$\alpha = \mathrm{i}\,(2\,Q_E + Q_i), \tag{18}$$

$$\beta = -Q_E\,(Q_E + Q_i). \tag{19}$$

Note that

$$\beta - \frac{1}{4}\,\alpha^2 = -Q_E\,(Q_E + Q_i) + \frac{1}{4}\,(2\,Q_E + Q_i)^2 = \frac{Q_i^{\,2}}{4}. \tag{20}$$

However (ERD: 5.3.34),

$$H(\hat{t}) = \mathrm{e}^{-\alpha\,\hat{t}/2}\,J_0[(\beta - \alpha^2/4)^{1/2}\,\hat{t}] \tag{21}$$

$$= \mathrm{e}^{-\mathrm{i}\,(Q_E + Q_i/2)\,\hat{t}}\,J_0\left(\frac{Q_i}{2}\,\hat{t}\right). \tag{22}$$

Thus,

$$\Psi_0(\hat{t}) = \frac{\Psi_\infty\,\epsilon}{\mathrm{i}\,(Q_E + Q_e)}\int_0^{\hat{t}}\left[1 - \mathrm{e}^{\,\mathrm{i}\,(Q_E + Q_e)\,(\hat{t}' - \hat{t})}\right] \times \mathrm{e}^{-\mathrm{i}\,(Q_E + Q_i/2)\,\hat{t}'}\,J_0\left(\frac{Q_i}{2}\,\hat{t}'\right)d\hat{t}'. \tag{23}$$

At small $\hat{t}$,

$$\Psi_0(\hat{t}) \simeq \Psi_\infty\,\epsilon\,\frac{\hat{t}^{\,2}}{\Gamma(3)}. \tag{24}$$

At large $\hat{t}$,

$$\Psi_0(\hat{t}) \simeq \frac{\Psi_\infty\,\epsilon}{\mathrm{i}\,(Q_E + Q_e)}\int_0^{\infty}\left[1 - \mathrm{e}^{\,\mathrm{i}\,(Q_E + Q_e)\,(\hat{t}' - \hat{t})}\right] \times \mathrm{e}^{-\mathrm{i}\,(Q_E + Q_i/2)\,\hat{t}'}\,J_0\left(\frac{Q_i}{2}\,\hat{t}'\right)d\hat{t}'. \tag{25}$$

But (GR:[2] 6.621.4)

$$\int_0^\infty \mathrm{e}^{-\alpha\,x}\,J_0(\beta\,x)\,dx = \frac{1}{\sqrt{\alpha^2+\beta^2}}, \tag{26}$$

so

$$\Psi_{b\,0}(\hat{t}) \simeq \frac{\Psi_\infty\,\epsilon}{\mathrm{i}\,(Q_E+Q_i)}\left\{\frac{1}{(\mathrm{i}\,Q_E)^{1/2}\,[\mathrm{i}\,(Q_E+Q_i)]^{1/2}} - \frac{\mathrm{e}^{-\mathrm{i}\,(Q_E+Q_e)\,\hat{t}}}{(\mathrm{i}\,Q_e)^{1/2}\,[\mathrm{i}\,(Q_e-Q_i)]^{1/2}}\right\}. \tag{27}$$

2.2 Viscous-Inertial Regime

In the Viscous-Inertial regime,

$$\hat{\Delta}_s(\hat{g}) = -\frac{\pi}{2}\,\frac{\Gamma(1/4)}{\Gamma(3/4)}\,\frac{1}{[\hat{g}+\mathrm{i}\,(Q_E+Q_e)]^{1/4}\,P_\varphi^{1/4}}, \tag{28}$$

$$F_s(\hat{g}) = -\frac{\pi^{1/2}}{2}\,\frac{\Gamma(1/4)}{\Gamma(3/4)}\,\frac{1}{[\hat{g}+\mathrm{i}\,(Q_E+Q_e)]^{3/2}\,P_\varphi^{1/2}}, \tag{29}$$

$$\frac{F_s(\hat{g})}{\hat{\Delta}_s(\hat{g})} = \frac{1}{\pi^{1/2}}\,\frac{1}{[\hat{g}+\mathrm{i}\,(Q_E+Q_e)]^{5/4}\,P_\varphi^{1/4}}. \tag{30}$$

Thus,

$$\bar{\Psi}_0(\hat{g}) = \frac{\Psi_\infty\,\epsilon}{\hat{g}}\,\frac{1}{[\hat{g}+\mathrm{i}\,(Q_E+Q_e)]^{5/4}}, \tag{31}$$

where

$$\epsilon = \frac{(-E_{ss})}{\pi^{1/2}\,S^{1/3}\,P_\varphi^{1/4}}. \tag{32}$$

In this case,

$$\bar{H}(\hat{g}) = \frac{1}{[\hat{g}+\mathrm{i}\,(Q_E+Q_e)]^{5/4}}, \tag{33}$$

so (ERD: 5.4.1)

$$H(\hat{t}) = \frac{1}{\Gamma(5/4)}\,\hat{t}^{\,1/4}\,\mathrm{e}^{-\mathrm{i}\,(Q_E+Q_e)\,\hat{t}}. \tag{34}$$

[2] I.S. Gradshteyn and I.M. Ryzhik, *Table of Integrals, Series, and Products*, Corrected and Enlarged Edition

Hence,

$$\Psi_0(\hat{t}) = \frac{\Psi_\infty\,\epsilon}{\Gamma(5/4)}\int_0^{\hat{t}} \hat{t}'^{\,1/4}\,{\rm e}^{-{\rm i}\,(Q_E+Q_e)\,\hat{t}'}\,d\hat{t}' \tag{35}$$

At small $\hat{t}$,

$$\Psi_0(\hat{t}) \simeq \Psi_\infty\,\epsilon\,\frac{\hat{t}^{\,5/4}}{\Gamma(9/4)}. \tag{36}$$

At large $\hat{t}$,

$$\Psi_0(\hat{t}) \simeq \frac{\Psi_\infty\,\epsilon}{\Gamma(5/4)}\int_0^{\infty} \hat{t}'^{\,1/4}\,{\rm e}^{-{\rm i}\,(Q_E+Q_e)\,\hat{t}'}\,d\hat{t}' \tag{37}$$

But (GR: 3.381.5),

$$\int_0^\infty x^{\nu-1}\,{\rm e}^{-{\rm i}\,q\,x}\,dx = \frac{\Gamma(\nu)}{({\rm i}\,q)^\nu}, \tag{38}$$

so

$$\Psi_0(\hat{t}) \simeq \frac{\Psi_\infty\,\epsilon}{[{\rm i}\,(Q_E+Q_e)]^{5/4}}. \tag{39}$$

2.3 Diffusive-Inertial Regime

In the Diffusive-Inertial regime,

$$\hat{\Delta}_s(\hat{g}) = -\frac{\pi\,D}{[\hat{g}+{\rm i}\,(Q_E+Q_e)]^{1/2}\,\iota_e^{1/2}\,P_\perp^{1/2}}, \tag{40}$$

$$F_s(\hat{g}) = -\frac{2\,D^2}{[\hat{g}+{\rm i}\,(Q_E+Q_e)]^2\,\iota_e\,P_\perp}, \tag{41}$$

$$\frac{F_s(\hat{g})}{\hat{\Delta}_s(\hat{g})} = \frac{2}{\pi}\,\frac{1}{[\hat{g}+{\rm i}\,(Q_E+Q_e)]^{3/2}\,\iota_e^{1/2}\,P_\perp^{1/2}\,D^{-1}}. \tag{42}$$

Thus,

$$\bar{\Psi}_0(\hat{g}) = \frac{\Psi_\infty\,\epsilon}{\hat{g}}\,\frac{1}{[\hat{g}+{\rm i}\,(Q_E+Q_e)]^{3/2}}, \tag{43}$$

where

$$\epsilon = \frac{2}{\pi}\,\frac{(-E_{ss})}{S^{1/3}\,\iota_e^{1/2}\,P_\perp^{1/2}\,D^{-1}}. \tag{44}$$

In this case,

$$\bar{H}(\hat{g}) = \frac{1}{[\hat{g}+{\rm i}\,(Q_E+Q_e)]^{3/2}}, \tag{45}$$

so (ERD: 5.4.1)

$$H(\hat{t}) = \frac{1}{\Gamma(3/2)}\,\hat{t}^{\,1/2}\,{\rm e}^{-{\rm i}\,(Q_E+Q_e)\,\hat{t}}. \tag{46}$$

Hence,

$$\Psi_0(\hat{t}) = \frac{\Psi_\infty\,\epsilon}{\Gamma(3/2)}\int_0^{\hat{t}}\hat{t}'^{\,1/2}\,{\rm e}^{-{\rm i}\,(Q_E+Q_e)\,\hat{t}'}\,d\hat{t}' \tag{47}$$

At small $\hat{t}$,

$$\Psi_0(\hat{t}) \simeq \Psi_\infty\,\epsilon\,\frac{\hat{t}^{\,3/2}}{\Gamma(5/2)}. \tag{48}$$

At large $\hat{t}$,

$$\Psi_0(\hat{t}) = \frac{\Psi_\infty\,\epsilon}{\Gamma(3/2)}\int_0^{\infty}\hat{t}'^{\,1/2}\,{\rm e}^{-{\rm i}\,(Q_E+Q_e)\,\hat{t}'}\,d\hat{t}' \tag{49}$$

But (GR: 3.381.5),

$$\int_0^\infty x^{\nu-1}\,{\rm e}^{-{\rm i}\,q\,x}\,dx = \frac{\Gamma(\nu)}{({\rm i}\,q)^\nu}, \tag{50}$$

so

$$\Psi_0(\hat{t}) \simeq \frac{\Psi_\infty\,\epsilon}{[{\rm i}\,(Q_E+Q_e)]^{3/2}}. \tag{51}$$

3 Constant-ψ Regimes

We shall not necessarily assume that $|S^{\,1/3}\,\hat{\Delta}_s| \gg |E_{ss}|$ in the constant-ψ regimes, which are characterized by $F_s(\hat{g}) = 1$, so

$$\bar{\Psi}_0(\hat{g}) \simeq E_{sb}\,\Psi_{b\,0}\,\frac{1}{\hat{g}}\,\frac{1}{S^{1/3}\,\hat{\Delta}_s(\hat{g}) - E_{ss}}. \tag{52}$$

3.1 Viscous-Resistive Regime

In the Viscous-Resistive regime,

$$\hat{\Delta}_s(\hat{g}) = \frac{6^{2/3}\,\pi\,\Gamma(5/6)}{\Gamma(1/6)}\,[\hat{g} + {\rm i}\,(Q_E+Q_e)]\,P_\varphi^{1/6}, \tag{53}$$

so

$$\bar{\Psi}_0(\hat{g}) = \frac{\Psi_\infty\,\epsilon}{\hat{g}}\,\frac{\epsilon}{[\hat{g} + {\rm i}\,(Q_E+Q_e)] + \epsilon}, \tag{54}$$

where

$$\epsilon = \frac{\Gamma(1/6)}{6^{3/2}\,\pi\,\Gamma(5/6)}\,\frac{(-E_{ss})}{S^{\,1/3}\,P_\varphi^{1/6}}. \tag{55}$$

Thus,

$$\bar{\Psi}_0(\hat{g}) = \frac{\Psi_\infty\,\epsilon}{\mathrm{i}\,(Q_E+Q_e)+\epsilon}\left(\frac{1}{\hat{g}} - \frac{1}{\hat{g}+[\mathrm{i}\,(Q_E+Q_e)+\epsilon]}\right), \tag{56}$$

giving

$$\Psi_0(\hat{t}) = \frac{\Psi_\infty\,\epsilon}{\mathrm{i}\,(Q_E+Q_e)+\epsilon}\left(1-\mathrm{e}^{-[\mathrm{i}\,(Q_E+Q_e)+\epsilon]\,\hat{t}}\right). \tag{57}$$

At small $\hat{t}$,

$$\Psi_0(\hat{t}) \simeq \Psi_\infty\,\epsilon\,\frac{\hat{t}}{\Gamma(2)}. \tag{58}$$

At large $\hat{t}$,

$$\Psi_0(\hat{t}) \simeq \frac{\Psi_\infty\,\epsilon}{\mathrm{i}\,(Q_E+Q_e)+\epsilon}. \tag{59}$$

3.2 Diffusive-Resistive Regime

In the Diffusive-Resistive regime,

$$\hat{\Delta}_s(\hat{g}) = \frac{2\pi\,\Gamma(3/4)}{\Gamma(1/4)}\,[\hat{g}+\mathrm{i}\,(Q_E+Q_e)]\,\iota_e^{1/4}\,P_\perp^{1/4}\,D^{-1/2}, \tag{60}$$

so

$$\bar{\Psi}_0(\hat{g}) = \frac{\Psi_\infty\,\epsilon}{\hat{g}}\,\frac{1}{[\hat{g}+\mathrm{i}\,(Q_E+Q_e)]+\epsilon}, \tag{61}$$

where

$$\epsilon = \frac{\Gamma(1/4)}{2\pi\,\Gamma(3/4)}\,\frac{(-E_{ss})}{S^{\,1/3}\,\iota_e^{1/4}\,P_\perp^{1/4}\,D^{-1/2}}. \tag{62}$$

Thus,

$$\bar{\Psi}_0(\hat{g}) = \frac{\Psi_\infty\,\epsilon}{\mathrm{i}\,(Q_E+Q_e)+\epsilon}\left(\frac{1}{\hat{g}} - \frac{1}{\hat{g}+[\mathrm{i}\,(Q_E+Q_e)+\epsilon]}\right), \tag{63}$$

giving

$$\mathit{\Psi}_0(\hat{t}) = \frac{\mathit{\Psi}_\infty\,\epsilon}{\mathrm{i}\,(Q_E+Q_e)+\epsilon}\left(1-\mathrm{e}^{-[\mathrm{i}\,(Q_E+Q_e)+\epsilon]\,\hat{t}}\right). \tag{64}$$

At small $\hat{t}$,

$$\mathit{\Psi}_0(\hat{t}) \simeq \mathit{\Psi}_\infty\,\epsilon\,\frac{\hat{t}}{\Gamma(2)}. \tag{65}$$

At large $\hat{t}$,

$$\mathit{\Psi}_0(\hat{t}) \simeq \frac{\mathit{\Psi}_\infty\,\epsilon}{\mathrm{i}\,(Q_E+Q_e)+\epsilon}. \tag{66}$$

3.3 Semi-Collisional Regime

In the Semi-Collisional regime,

$$\hat{\mathit{\Delta}}_s(\hat{g}) = \pi\,[\hat{g}+\mathrm{i}\,(Q_E+Q_e)]\,(\hat{g}+\mathrm{i}\,Q_E)^{1/2}\,D^{-1}, \tag{67}$$

so

$$\bar{\mathit{\Psi}}_0(\hat{g}) = \frac{\mathit{\Psi}_\infty\,\epsilon}{\hat{g}}\,\frac{1}{[\hat{g}+\mathrm{i}\,(Q_E+Q_e)]\,(\hat{g}+\mathrm{i}\,Q_E)^{1/2}+\epsilon}, \tag{68}$$

where

$$\epsilon = \frac{1}{\pi}\,\frac{(-E_{ss})}{S^{\,1/3}\,D^{-1}}. \tag{69}$$

If we neglect ϵ in the denominator then

$$\bar{\mathit{\Psi}}_0(\hat{g}) \simeq \frac{\mathit{\Psi}_\infty\,\epsilon}{\hat{g}}\,\frac{1}{[\hat{g}+\mathrm{i}\,(Q_E+Q_e)]\,(\hat{g}+\mathrm{i}\,Q_E)^{1/2}}, \tag{70}$$

giving

$$\bar{\mathit{\Psi}}_0(\hat{g}) \simeq \frac{\mathit{\Psi}_\infty\,\epsilon}{\mathrm{i}\,(Q_E+Q_e)}\left[\frac{1}{\hat{g}}-\frac{1}{\hat{g}+\mathrm{i}\,(Q_E+Q_e)}\right]\frac{1}{(\hat{g}+\mathrm{i}\,Q_E)^{1/2}}. \tag{71}$$

In this case,

$$\bar{H}(\hat{g}) = \frac{1}{(\hat{g}+\mathrm{i}\,Q_E)^{1/2}}, \tag{72}$$

so (ERD: 5.4.1)

$$H(\hat{t}) = \frac{1}{\Gamma(1/2)}\,\hat{t}^{-1/2}\,\mathrm{e}^{-\mathrm{i}\,Q_E\,\hat{t}}. \tag{73}$$

Hence,

$$\begin{aligned}\Psi_0(\hat{t}) = {}&\frac{\Psi_\infty\,\epsilon}{\Gamma(1/2)\,\mathrm{i}\,(Q_E+Q_e)}\\ &\times\int_0^{\hat{t}}\left[1-\mathrm{e}^{\,\mathrm{i}\,(Q_E+Q_e)\,(\hat{t}'-\hat{t})}\right]\hat{t}'^{-1/2}\,\mathrm{e}^{-\mathrm{i}\,Q_E\,\hat{t}'}\,d\hat{t}'.\end{aligned} \tag{74}$$

At small $\hat{t}$,

$$\Psi_0(\hat{t}) = \Psi_\infty\,\epsilon\,\frac{\hat{t}^{\,3/2}}{\Gamma(5/2)}. \tag{75}$$

At large $\hat{t}$,

$$\begin{aligned}\Psi_0(\hat{t}) = {}&\frac{\Psi_\infty\,\epsilon}{\Gamma(1/2)\,\mathrm{i}\,(Q_E+Q_e)}\\ &\times\int_0^{\infty}\left[1-\mathrm{e}^{\,\mathrm{i}\,(Q_E+Q_e)\,(\hat{t}'-\hat{t})}\right]\hat{t}'^{-1/2}\,\mathrm{e}^{-\mathrm{i}\,Q_E\,\hat{t}'}\,d\hat{t}'.\end{aligned} \tag{76}$$

But (GR: 3.381.5),

$$\int_0^\infty x^{\nu-1}\,\mathrm{e}^{-\mathrm{i}\,q\,x}\,dx = \frac{\Gamma(\nu)}{(\mathrm{i}\,q)^\nu}, \tag{77}$$

It follows that

$$\Psi_0(\hat{t}) = \frac{\Psi_\infty\,\epsilon}{\mathrm{i}\,(Q_E+Q_e)}\left[\frac{1}{(\mathrm{i}\,Q_E)^{1/2}}-\frac{\mathrm{e}^{-\mathrm{i}\,(Q_E+Q_e)\,\hat{t}}}{(-\mathrm{i}\,Q_e)^{1/2}}\right]. \tag{78}$$

3.4 Resistive-Inertial Regime

In the Resistive-Inertial regime,

$$\hat{\Delta}_s(\hat{g}) = \frac{2\pi\,\Gamma(3/4)}{\Gamma(1/4)}\,[\hat{g}+\mathrm{i}\,(Q_E+Q_e)]^{3/4}\,(\hat{g}+\mathrm{i}\,Q_E)^{1/4}\,[\hat{g}+\mathrm{i}\,(Q_E+Q_i)]^{1/4}, \tag{79}$$

so

$$\bar{\Psi}_0(\hat{g}) = \frac{\Psi_\infty\,\epsilon}{\hat{g}}\,\frac{1}{[\hat{g}+\mathrm{i}\,(Q_E+Q_e)]^{3/4}\,(\hat{g}+\mathrm{i}\,Q_E)^{1/4}\,[\hat{g}+\mathrm{i}\,(Q_E+Q_i)]^{1/4}+\epsilon}, \tag{80}$$

where

$$\epsilon = \frac{\Gamma(1/4)}{2\pi\,\Gamma(3/4)}\,\frac{(-E_{ss})}{S^{\,1/3}}. \tag{81}$$

If we neglect ϵ in the denominator then

$$\bar{\Psi}_0(\hat{g}) \simeq \frac{\Psi_\infty\,\epsilon}{\hat{g}}\,\frac{1}{[\hat{g}+{\rm i}\,(Q_E+Q_e)]^{3/4}\,(\hat{g}+{\rm i}\,Q_E)^{1/4}\,[\hat{g}+{\rm i}\,(Q_E+Q_i)]^{1/4}}. \tag{82}$$

Now, if

$$\bar{\Psi}_0(\hat{g}) \simeq \Psi_\infty\,\epsilon\,\frac{H(\hat{g})}{\hat{g}} \tag{83}$$

then

$$\Psi_0(\hat{t}) = \Psi_\infty\,\epsilon\int_0^{\hat{t}} H(\hat{t}')\,d\hat{t}'. \tag{84}$$

Furthermore, if

$$\bar{H}(\hat{g}) = \bar{H}_1(\hat{g})\,\bar{H}_2(\hat{g}) \tag{85}$$

then

$$H(\hat{t}) = \int_0^{\hat{t}} H_1(\hat{t}')\,H_2(\hat{t}-\hat{t}')\,d\hat{t}'. \tag{86}$$

It follows that

$$\Psi_0(\hat{t}) = \Psi_\infty\,\epsilon\int_0^{\hat{t}}\left[\int_0^{\hat{t}'} H_1(\hat{t}'')\,H_2(\hat{t}'-\hat{t}'')\,d\hat{t}''\right]d\hat{t}'. \tag{87}$$

In the present case

$$\bar{H}_1(\hat{g}) = \frac{1}{(\hat{g}+{\rm i}\,Q_E)^{1/4}\,[\hat{g}+{\rm i}\,(Q_E+Q_i)]^{1/4}}, \tag{88}$$

$$\bar{H}_2(\hat{g}) = \frac{1}{[\hat{g}+{\rm i}\,(Q_E+Q_e)]^{3/4}}, \tag{89}$$

so (ERD: 5.4.5, AS:[3] 9.6.3, ERD: 5.4.1)

$$H_1(\hat{t}) = \frac{\pi^{1/2}}{\Gamma(1/4)}\,Q_i^{1/4}\,\hat{t}^{-1/4}\,{\rm e}^{-{\rm i}\,(Q_E+Q_i/2)\,\hat{t}}\,J_{-1/4}\!\left(\frac{Q_i}{2}\,\hat{t}\right), \tag{90}$$

[3]M. Abramowitz and I.A. Stegun, *Handbook of Mathematical Functions* (Dover, New York NY, 1964).

$$H_2(\hat{t}) = \frac{1}{\Gamma(3/4)}\,\hat{t}^{-1/4}\,\mathrm{e}^{-\mathrm{i}\,(Q_E+Q_e)\,\hat{t}}. \tag{91}$$

Note that

$$\Gamma(1/4)\,\Gamma(3/4) = \sqrt{2}\,\pi. \tag{92}$$

Thus,

$$\Psi_0(\hat{t}) = \frac{\Psi_\infty\,\epsilon\,Q_i^{1/4}}{\sqrt{2\pi}}\int_0^{\hat{t}}\mathrm{e}^{-\mathrm{i}\,(Q_E+Q_e)\,\hat{t}'}\left[\int_0^{\hat{t}'}(\hat{t}'-\hat{t}'')^{-1/4}\,\hat{t}''^{-1/4}\right.$$
$$\left.\times\mathrm{e}^{\mathrm{i}\,(Q_e-Q_i/2)\,\hat{t}''}\,J_{-1/4}\!\left(\frac{Q_i}{2}\,\hat{t}''\right)d\hat{t}''\right]d\hat{t}'. \tag{93}$$

At small $\hat{t}$ (AS: 9.1.10),

$$J_{-1/4}\!\left(\frac{Q_i}{2}\,\hat{t}\right) \simeq \frac{\sqrt{2}\,Q_i^{-1/4}\,\hat{t}^{-1/4}}{\Gamma(3/4)}, \tag{94}$$

so

$$\Psi_0(\hat{t}) \simeq \frac{\Psi_\infty\,\epsilon}{\sqrt{\pi}\,\Gamma(3/4)}\int_0^{\hat{t}}\left[\int_0^{\hat{t}'}(\hat{t}'-\hat{t}'')^{-1/4}\,\hat{t}''^{-1/2}\,d\hat{t}''\right]d\hat{t}'. \tag{95}$$

But,

$$\int_0^{\hat{t}'}(\hat{t}'-\hat{t}'')^{-1/4}\,\hat{t}''^{-1/2}\,d\hat{t}'' = \hat{t}'^{\,1/4}\int_0^1(1-x)^{-1/4}\,x^{-1/2}\,dx, \tag{96}$$

and (AS: 6.2.1, 6.2.2)

$$\int_0^1(1-x)^{-1/4}\,x^{-1/2}\,dx = \frac{\Gamma(1/2)\,\Gamma(3/4)}{\Gamma(5/4)}, \tag{97}$$

so

$$\Psi_0(\hat{t}) \simeq \Psi_\infty\,\epsilon\,\frac{\hat{t}^{\,5/4}}{\Gamma(9/4)}. \tag{98}$$

At large $\hat{t}$,

$$\Psi_0(\hat{t}) \simeq \frac{\Psi_\infty\,\epsilon\,Q_i^{1/4}}{\sqrt{2\pi}}\int_0^{\infty}\mathrm{e}^{-\mathrm{i}\,(Q_E+Q_e)\,\hat{t}'}\left[\int_0^{\hat{t}'}(\hat{t}'-\hat{t}'')^{-1/4}\,\hat{t}''^{-1/4}\right.$$
$$\left.\times\mathrm{e}^{\mathrm{i}\,(Q_e-Q_i/2)\,\hat{t}''}\,J_{-1/4}\!\left(\frac{Q_i}{2}\,\hat{t}''\right)d\hat{t}''\right]d\hat{t}', \tag{99}$$

so

$$\Psi_0(\hat{t}) \simeq \frac{\Psi_\infty\,\epsilon\,Q_i^{1/4}}{\sqrt{2\pi}}\int_0^\infty\left[\int_{\hat{t}''}^\infty (\hat{t}'-\hat{t}'')^{-1/4}\,{\rm e}^{-{\rm i}\,(Q_E+Q_e)\,\hat{t}'}\,d\hat{t}'\right]$$
$$\times\,\hat{t}''^{-1/4}\,{\rm e}^{{\rm i}\,(Q_e-Q_i/2)\,\hat{t}''}\,J_{-1/4}\left(\frac{Q_i}{2}\,\hat{t}''\right)\,d\hat{t}''. \tag{100}$$

But,

$$\int_{\hat{t}''}^\infty (\hat{t}'-\hat{t}'')^{-1/4}\,{\rm e}^{-{\rm i}\,(Q_E+Q_e)\,\hat{t}'}\,d\hat{t}' = {\rm e}^{-{\rm i}\,(Q_E+Q_e)\,\hat{t}''}\int_0^\infty x^{-1/4}\,{\rm e}^{-{\rm i}\,(Q_E+Q_e)\,x}\,dx, \tag{101}$$

and (GR: 3.381.5),

$$\int_0^\infty x^{\nu-1}\,{\rm e}^{-{\rm i}\,q\,x}\,dx = \frac{\Gamma(\nu)}{({\rm i}\,q)^\nu}, \tag{102}$$

so

$$\int_{\hat{t}''}^\infty (\hat{t}'-\hat{t}'')^{-1/4}\,{\rm e}^{-{\rm i}\,(Q_E+Q_e)\,\hat{t}'}\,d\hat{t}' = {\rm e}^{-{\rm i}\,(Q_E+Q_e)\,\hat{t}''}\,\frac{\Gamma(3/4)}{[{\rm i}\,(Q_E+Q_e)]^{3/4}}. \tag{103}$$

It follows that

$$\Psi_0(\hat{t}) = \frac{\Psi_\infty\,\epsilon\,Q_i^{1/4}}{\sqrt{2\pi}}\,\frac{\Gamma(3/4)}{[{\rm i}\,(Q_E+Q_e)]^{3/4}}$$
$$\times\int_0^\infty \hat{t}''^{-1/4}\,{\rm e}^{-{\rm i}\,(Q_E+Q_i/2)\,\hat{t}''}\,J_{-1/4}\left(\frac{Q_i}{2}\,\hat{t}''\right)\,d\hat{t}''. \tag{104}$$

But (GR: 6.623.1),

$$\int_0^\infty \hat{t}''^{-1/4}\,{\rm e}^{-{\rm i}\,(Q_E+Q_i/2)\,\hat{t}''}\,J_{-1/4}\left(\frac{Q_i}{2}\,\hat{t}''\right)\,d\hat{t}''$$
$$= \frac{(Q_i)^{-1/4}\,\Gamma(1/4)}{\sqrt{\pi}\,({\rm i}\,Q_E)^{1/4}\,[{\rm i}\,(Q_E+Q_i)]^{1/4}}. \tag{105}$$

Now (AS: 6.1.17),

$$\Gamma(3/4)\,\Gamma(1/4) = \sqrt{2}\,\pi, \tag{106}$$

so

$$\Psi_0(\hat{t}) \simeq \frac{\Psi_\infty\,\epsilon}{[{\rm i}\,(Q_E+Q_e)]^{3/4}\,({\rm i}\,Q_E)^{1/4}\,[{\rm i}\,(Q_E+Q_i)]^{1/4}}. \tag{107}$$